

Image Preprocessing:

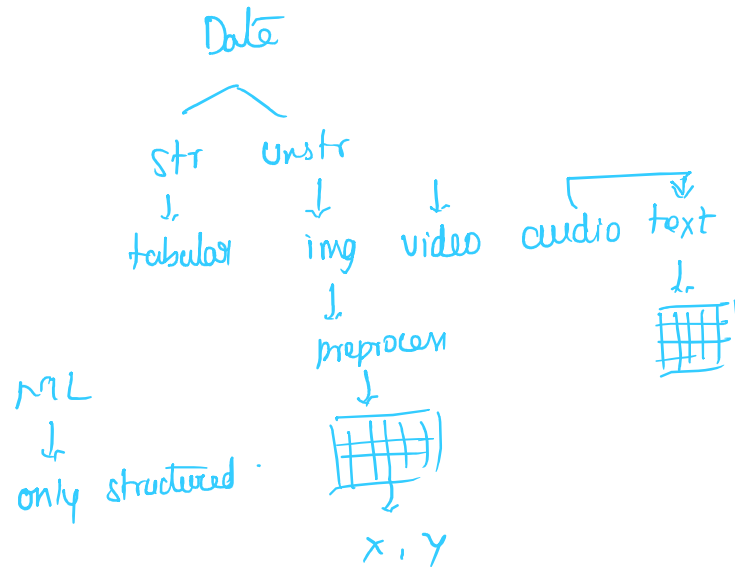
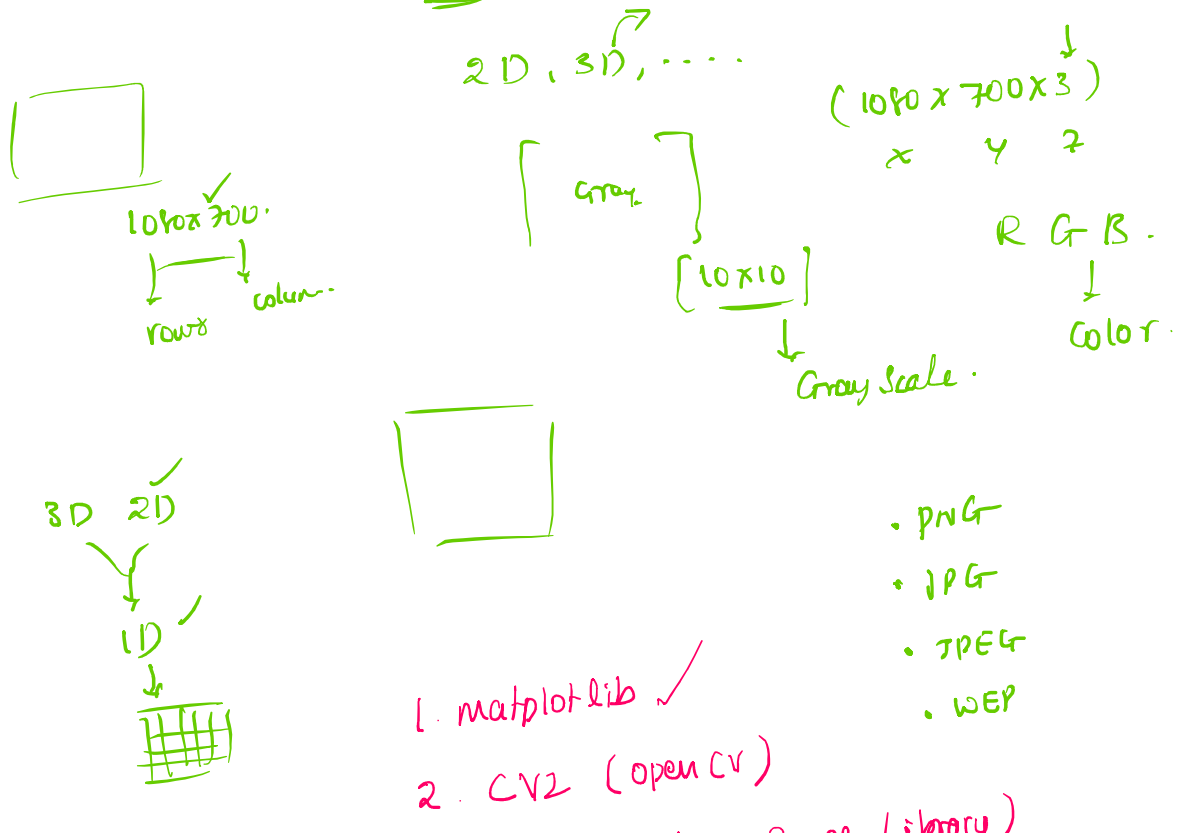


image → collection of pixels.

Numpy:



###

2. CV2 (OpenCV)

3. PIL (Python Image Library)

```
img = plt.imread("file.png")
```

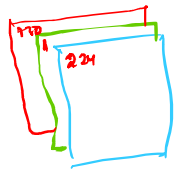
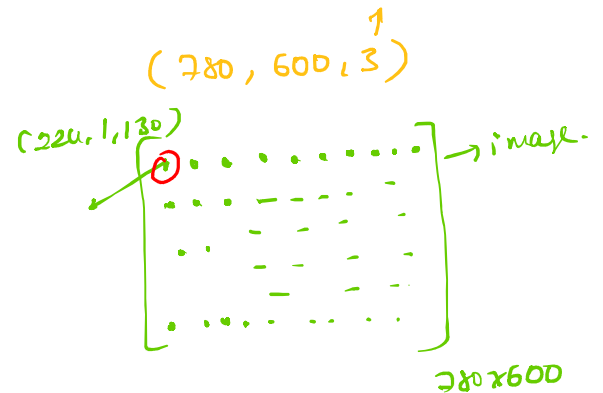
```
print(img)
```

↳ array.

R G B
[[[224, 1, 130] ✓
= [120, 3, 15]
:
]]]

img.shape

img.ndim



R G B

0-255

(0,0,0) → Black

(255,255,255) → white

(255,0,0) → Red

(0,255,0) → Green

(0,0,255) → Blue

100x100

↓

10000

↳ 10MP

1000x1000

40

↓

2048x3600

1MP → 1 Million Pixel

↓

780x600

import PIL

!pip install pillow

from PIL import Image

img = Image.open("file.png")

img.

~~img~~ = np.array(img) ✓

img_arr.shape

img.mode ✓

RGB

PIL → RGB

PIL → RGB ←

CV2 → BGR

3D → 2D

color → Gray

[120, 13, 120] → one single number

↓ mean

gray = img.convert("L")

g_img = np.array(——).shape

~~mode~~

g_img.mode

L → Luma Transformation

[120, 13, 120]

↓ single num.

(mean)

↓ weight avg ✓

$$\frac{A+B+C}{3}$$

$$\frac{1}{3} \times 120 + \frac{1}{3} \times 13 + \frac{1}{3} \times 120$$

$$\frac{A}{3} + \frac{B}{3} + \frac{C}{3}$$

~~2~~ 21

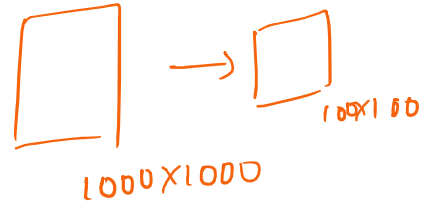
↑

$$\frac{R \times 299}{1000} + \frac{G \times 587}{1000} + \frac{B \times 114}{1000} \Rightarrow 215$$

$$\frac{R \times 333}{1000} + \frac{G \times 333}{1000} + \frac{B \times 333}{1000}$$

↓ ↓
[1, 2, 3]
[5, 2, 9]
[3, 5, 9]

[::, ::, 0]



(21-1)

$$\begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix} \rightarrow 4 \times 4$$

2x2

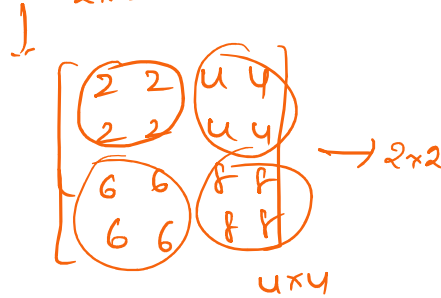
Interpolation
↓

400x600

2,40,000

100x120
→ 1,20,000

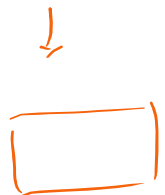
2x2



maxpool

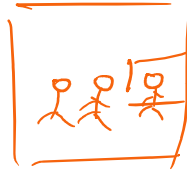
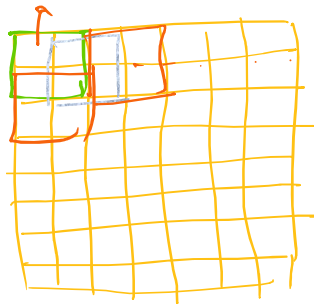
$$\begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 7 & 8 & 9 & 10 & ? \\ - & - & - & - & - \\ - & - & - & - & - \end{bmatrix}$$



1000 images
↓
1000x1000

Broad



(b) (c)

3D → 2D → 1D



CNN
YOLO



↓
Dimension
similar
since
D
S
S
D

Data augmentation :
✓

- Duplicate data
- Synthetic data

30
↓
300

